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In-flight positron annihilation as a probe of feebly interacting particles

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Core-collapse supernovae (SNe) provide a unique environment to study Feebly Interacting Particles (FIPs) such as Axion-Like Particles (ALPs), sterile neutrinos, and Dark Photons (DPs). This paper focuses on heavy FIPs produced in SNe, whose decay produces electrons and positrons, generating observable secondary signals during their propagation and annihilation. We focus on the In-flight Annihilation (IA) of positrons, which emerge as the most significant contribution to the resulting γ-ray spectrum. Using data from COMPTEL and EGRET we derive the most stringent bounds on the FIP-electron couplings for heavy ALPs, sterile neutrinos, and DPs. These results strengthen existing bounds by one to two orders of magnitude, depending on the FIP model.

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I. INTRODUCTION

Core-collapse supernovae (SNe) are commonly recognized as nature's most powerful factories for light and weakly interacting particles. Therefore, they can be used to investigate neutrino physics [1–3] and the existence of exotic Feebly Interacting Particles (FIPs) [1, 4]. FIPs include, among others, axions and Axion-Like Particles (ALPs) [5–10], scalar bosons [11], sterile neutrinos [12–16], Dark Photons (DPs) [17, 18], light CP-even scalars [19, 20], dark flavored particles [21] and unparticles [22]. All of these FIPs are characterized by the common feature of having extremely suppressed interactions with the Standard Model (SM) particles, making their direct detection extremely challenging [23]. Therefore, SNe are often the most favorable environments to probe FIPs, lying at the frontier of low-energy and high-intensity physics.

Notably, heavy FIPs in a SN core might be able to escape the stellar volume decaying into SM particles along their path in the interstellar medium. Therefore, the investigation of observable astrophysical fluxes in coincidence with SN events is a promising opportunity to probe FIPs and their properties. An interesting research line is linked to the study of electrophilic FIPs more massive than ~ 1 MeV and decaying into electron-positron pairs [15, 24–28]. Considering Galactic SNe, the propagation and annihilation of the electrons and positrons injected by FIP decays, gives rise to a diffuse signal spanning from X-ray to γ-ray energies. In particular, Inverse-Compton (IC) scattering of leptons on diffuse photon fluxes, generates a sub-MeV flux detectable in X-ray Multi-Mirror Mission (XMM-Newton) [29] and low-energy Spectrometer on INTEGRAL (SPI) data [30–33]. At 511 keV, the electron-positron annihilation line accurately measured by SPI, gives stringent information on the positron injection. In the γ-ray range, bremsstrahlung emission produces a sizable signal that can be probed by SPI high-energy data, the Imaging Compton Telescope (COMPTEL) [34] and the Energetic Gamma Ray Experiment Telescope (EGRET) [35]. Moreover, it was recently realized that In-flight positron Annihilation (IA) [36] is the most important contribution to the γ-ray signal [24]. Moreover, since the primary lepton flux injected by FIP decays is measurable by Voyager-1 [37], it can be interpreted as a sensitive probe of new physics as well as photon observables.

In this work, we analyze several FIP models, from the production in SNe, to the induced secondary photon fluxes. We motivate that, in general, the IA signal is the most constraining probe of the FIP-electron coupling. Therefore, we evaluate the IA bound for ALPs, sterile neutrinos and DPs. We structure this paper as follows: in Sec. II, we describe the case of SNe as sources of FIPs injecting positrons in the interstellar medium (ISM), in Sec. III we present our results by comparing the secondary γ-ray production with data and finally in Sec. IV we conclude.

II. POSITRON SOURCES: THE CASE OF FEEBLY INTERACTING PARTICLES

In our previous paper [24] we made use of a very generic parametrization of the $e^{\pm}$ emission via FIP decay, $X \to$